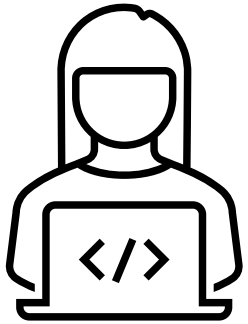
Welcome :-D

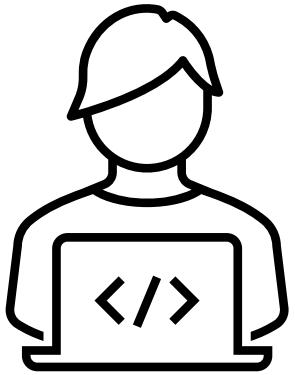
Agenda

- 01:00pm – 01:30pm Sample Collection Sheet, Code, and Miscellaneous
- 01:30pm - 2:20pm Introduction to Coding and Principal Component Analysis (PCA)
- 2:30pm – 4:00pm Hands-on Coding Tutorial

Rules

- Please interrupt to ask questions at any time!
- The workshop will be recorded, and the second part will be shared on YouTube.





Sample Collection Sheet, the motivation!

Imagine one of you gets offered a dream job, quits the PhD, and disappears tomorrow. Someone else is asked to finish the project. All they have are the SCIEX data files from the processing station. Where do they even start?

<u>The sample collection sheet aims to:</u>

- Collect the minimum information needed to interpret targeted PFAS data.
- Being able to interpret and use all data allows us to
    - Calculate intervals for EIS recoveries per matrix type, as well as default values for MDLs
    - Easily troubleshoot problems with standards
    - Create maps of PFAS point sources and matrix types
    - ....

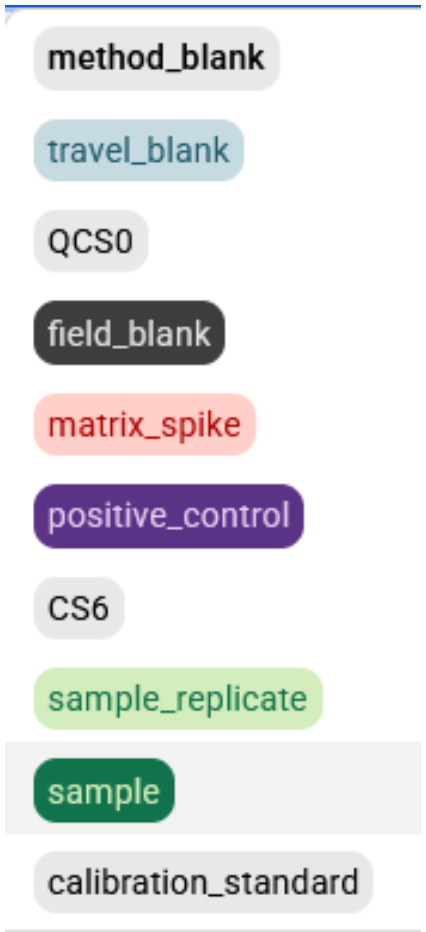
Sample Names – Naming Conventions

- Avoid white spaces
- Exclude special characters: & ,* % # ;*)(!@$^~' { [?<-
- 32 characters or less.
- What **information is important** about your files and **makes each file distinct**?
- The sample names will be used within SCIEX and will likely carry through into your first paper draft, so it's worth spending an extra 30 seconds choosing them carefully.

PBS_Receiver_0min_2
Vehicle_Receiver_0min_1
Vehicle_Receiver_0min_2
PBS_Donor_0min_1
PBS_Donor_0min_2
Vehicle_Donor_0min_1
Vehicle_Donor_0min_2
PBS_Donor_30min_1
PBS_Donor_30min_2
Vehicle_Donor_30min_1
Vehicle_Donor_30min_2
PBS_Receiver_30min_1
PBS_Receiver_30min_2
Vehicle_Receiver_30min_1
Vehicle_Receiver_30min_2
Cell_Growth_Plate_0min
Cell_Growth_Plate_30min

Sample Types and Definitions.

- **Sample replicates:** multiple, independent samples taken and processed under identical conditions to measure variability.
- **Matrix spike**: samples spiked with 4 ng of native standards prior to extraction and cleanup.
- **Positive control**: more generally defined as any type of positive control needed for individual experiments. In this case, details such as which part of the matrix is analyzed (e.g. matrix without organisms for cells or plankton or without fiber for sorption experiments), and which spiking solution is used are left to the researchers. For these sample types it is recommended to use the "sample comment" column to add details.



Matrix Types

serum

radiello

oysters

DGT

PE_tube

SPME

PUF

fish

egg

liver

plasma

Other

cells

cell media

saline

direct_injection

extracted_water

sediment

vegetation

Mass/Volume/Number and units.

- You want to calculate PFAS concentrations?
- We detect **ng per sample.** How do you get ng/mL, ng/ng, etc?
- You divide by the sample volume or weight (depending on matrix type). You need to know the mass or volume of your environmental samples before extraction.
- Without this information the data is worthless. So please track it in the data collection sheet.
- For QAQC samples the volume/mass defaults to 1 sample.

plasma	0.015	mL
plasma	0.015	mL
extracted_water	5	mL
extracted_water	75	mL

EIS and NIS volume and number

If there are problems with certain standards and you are affected, we want you to know!

We only know you cleanly log your standards in the sample book.

Please log cleanly log which standards you spiked in the sample book!

The Code...

- Automatically processes your SCIEX data and performs QAQC
- Is heavily validated within Julie's liver work
- Automatically merges methods (core/extended) and sample batches.
- Can be used to save your time and doublecheck your calculations.

Sample Number	5:3 FTCA	7:3 FTCA	FOEA	PFPrS	PFBS	PFPeS
PFAS CS6 4ng/ml	0.01	0.01	-0.01	0.14	0.00	0.26
PFAS QCS0 0ng/ml						
PFAS QCS0 0ng/ml						
Groundwater #1	-0.01				0.00	0.35
Groundwater #2				-0.02		0.31
Groundwater #3	-0.02	0.01			0.00	0.33
Groundwater #4	-0.01				0.00	0.38
Groundwater #5	0.01			-0.02		0.38
Surfacewater #1	0.00	0.01		0.52	0.02	0.68
Surfacewater #2	0.01				0.00	0.64
Surfacewater #3				0.29	0.00	0.31
Surfacewater #4		0.01		0.31	0.00	
Surfacewater #5				0.11	0.00	0.33

Retention Time Differences

Sample Number	5:3 FTCA	7:3 FTCA	FOEA	PFPrS	PFBS	PFPeS
PFAS CS6 4ng/ml	4.42	2.17	4.00	3.57	3.38	3.91
PFAS QCS0 0ng/ml	0.00	0.00	0.00	0.00	0.00	0.00
PFAS QCS0 0ng/ml	0.00	0.00	0.00	0.00	0.00	0.00
Groundwater #1	0.00	0.00	0.00	0.00	0.15	0.06
Groundwater #2	0.00	0.00	0.00	0.08	0.00	0.04
Groundwater #3	0.02	0.00	0.00	0.00	0.08	0.06
Groundwater #4	0.02	0.00	0.00	0.00	3.75	0.06
Groundwater #5	0.03	0.00	0.00	0.04	0.00	0.10
Surfacewater #1	0.31	0.03	0.00	0.08	3.33	0.22
Surfacewater #2	0.02	0.00	0.00	0.00	0.41	0.01
Surfacewater #3	0.00	0.00	0.00	0.01	0.28	0.05
Surfacewater #4	0.00	0.01	0.00	0.42	0.40	0.00
Surfacewater #5	0.00	0.00	0.00	0.29	0.09	0.06

Method Detection Limits

Sample Number	5:3 FTCA	7:3 FTCA	FOEA	PFPrS	PFBS	PFPeS
PFAS CS6 4ng/ml	74	47	47	88	83	83
PFAS QCS0 0ng/ml	74	42	42	87	90	90
PFAS QCS0 0ng/ml	81	56	56	84	95	95
Groundwater #1	74	57	57	89	92	92
Groundwater #2	56	51	51	83	80	80
Groundwater #3	60	55	55	90	89	89
Groundwater #4	69	48	48	82	80	80
Groundwater #5	69	47	47	81	84	84
Surfacewater #1	70	47	47	114	116	116
Surfacewater #2	64	55	55	96	100	100
Surfacewater #3	68	48	48	84	85	85
Surfacewater #4	69	71	71	98	88	88
Surfacewater #5	65	66	66	88	86	86

EIS-Recovery Rates

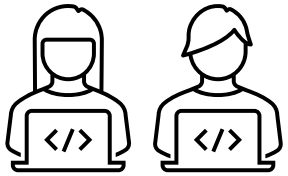
Sample Number	5:3 FTCA	7:3 FTCA	FOEA	PFPrS	PFBS	PFPeS
PFAS CS6 4ng/ml	20	11	-93	-6	1	13
PFAS QCS0 0ng/ml						
PFAS QCS0 0ng/ml						
Groundwater #1	inf		inf	inf	-2	inf
Groundwater #2	inf	inf		inf	inf	inf
Groundwater #3	-22	inf	inf	inf	1	inf
Groundwater #4	-46	inf	inf	inf	5	inf
Groundwater #5	inf		inf	-98	inf	inf
Surfacewater #1	inf	inf		-97	-27	inf
Surfacewater #2	inf	inf	inf	inf	10	-81
Surfacewater #3		inf		-83	10	-95
Surfacewater #4		inf		-99	25	
Surfacewater #5				-99	-5	inf

Ion-Abundance Ratio Deviations

DOI:
https://doi.org/10.5281/zenodo.17794240

Questions?

What is Code?



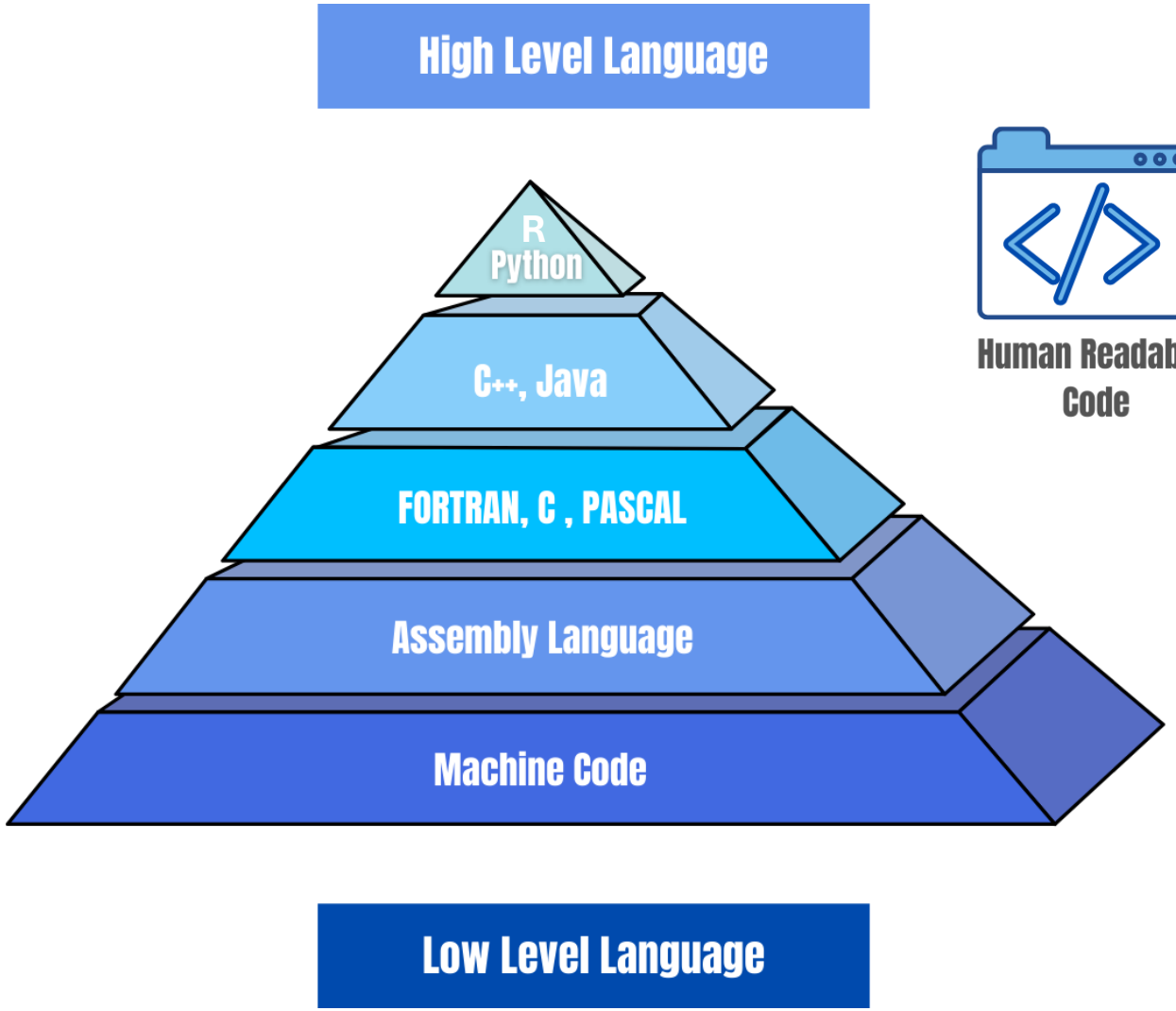
- Code is a set of precise, logical instructions. It is setup like a recipe and executes commands from top-to-bottom, one instruction at a time.
- Code serves as a translator between human intent and computer binary (0s and 1s), allowing computers to manipulate data, automate processes, and run applications

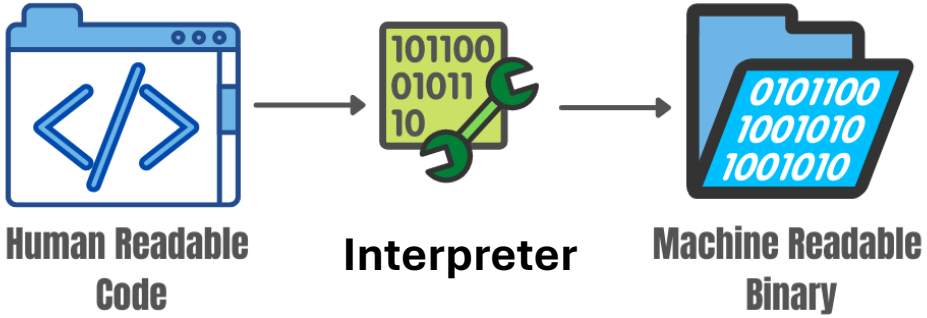
[1]
✓ 0s

```
number_of_participants <- 15
number_of_donuts <- 12
donuts_per_person <- number_of_donuts / number_of_participants
print("Hello PFAS explorers! You get x donuts:")
print(donuts_per_person)
```

```
[1] "Hello PFAS explorers! You get x donuts:"
[1] 0.8
```

What is a Coding Language?

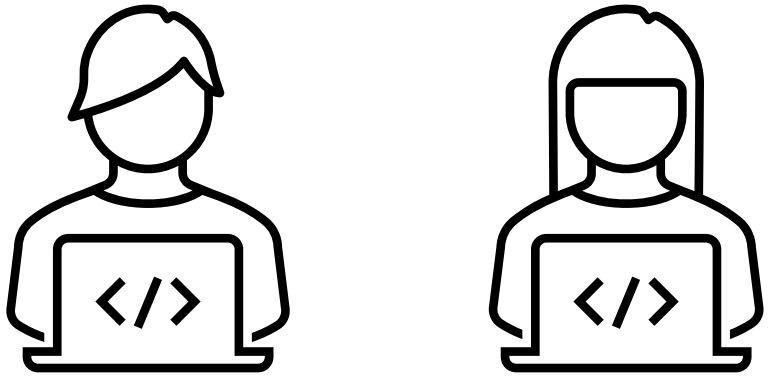




Your computer does not "understand" R or Python by itself. It needs an **interpreter** that reads your code line-by-line, translates it to machine code and executes it.

https://www.batoi.com/blogs/developers/the-language-machines-612cd9b994281

How to run Code?



- Integrated Development Environments (IDEs) provide a code editor, a run button, a console connected to the interpreter, and tools for plotting, debugging, and file browsing.
- IDE facilitates writing and running Code as well as viewing results.

Language	Level	Interpreter & IDE	comment
Jupyter notebook (Python)	Beginner	google colab, jupyter.org	- web-based, no installation required - file management can be annoying - data security concerns
R		webr.sh	
Python	Expert	Anaconda + VSCode	- Anaconda (package management and interpreter) + VSCode (IDE)
R	Expert	R + RStudio	- R (interpreter) + R Studio Code (IDE)

Mini exercise

Open a python, jupyter notebook or R IDE on your computer.

[1]
0s

```
number_of_participants <- 15
number_of_donuts <- 12
donuts_per_person <- number_of_donuts / number_of_participants
print("Hello PFAS explorers! You get x donuts:")
print(donuts_per_person)
```

```
[1] "Hello PFAS explorers! You get x donuts:"
[1] 0.8
```

Calculate and print the actual number of donuts per person for today.
What happens if you change the order of commands?

DEMO

Python group and R group

Open webr.sh.

Point out file handling and make sure people know where file is saved

Show example in R

Show what happens when you change order of variables

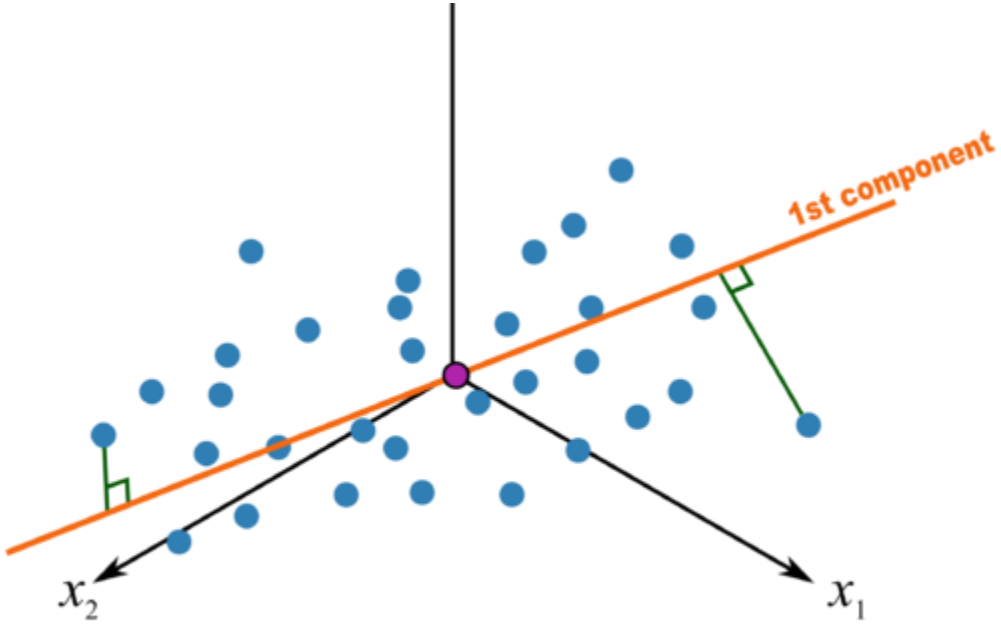
Find someone who wants to share it on jupyter.

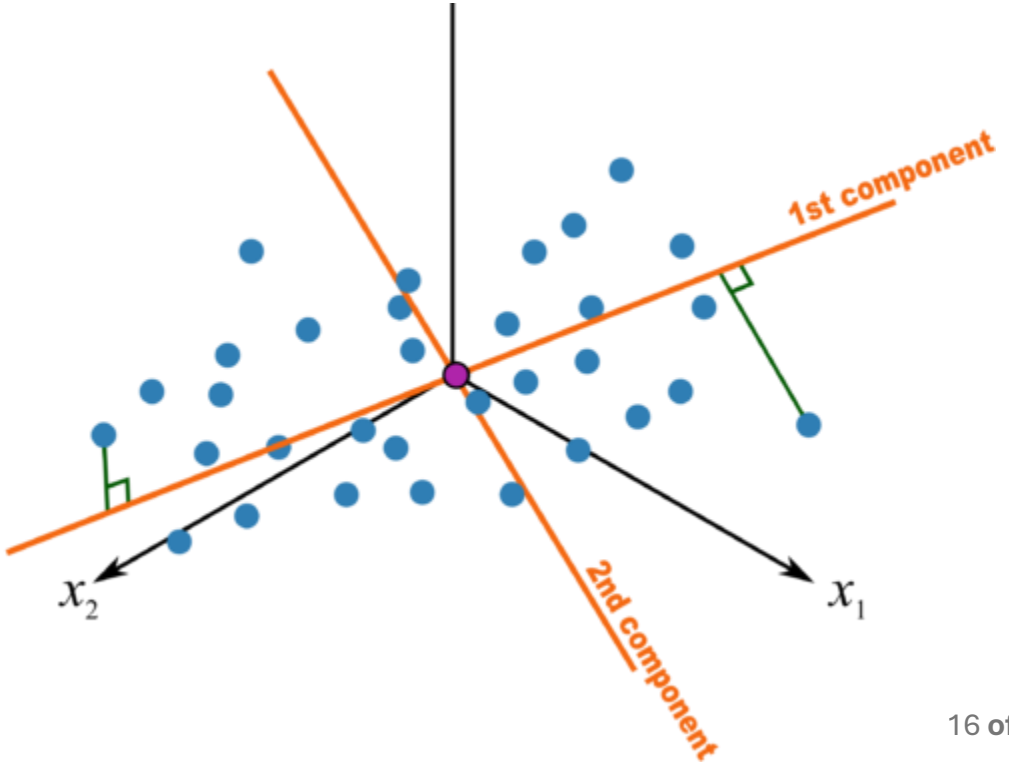
Coding Tipps

- Run Code in small steps.
- Read (the first lines) of error messages.
- Use meaningful variable names.
- Define parameters as variables at the top, don’t use numbers directly in calculations.
- Learn how to use functions and loops and avoid copying and pasting code too often.

What is Principal Component Analysis (PCA)?

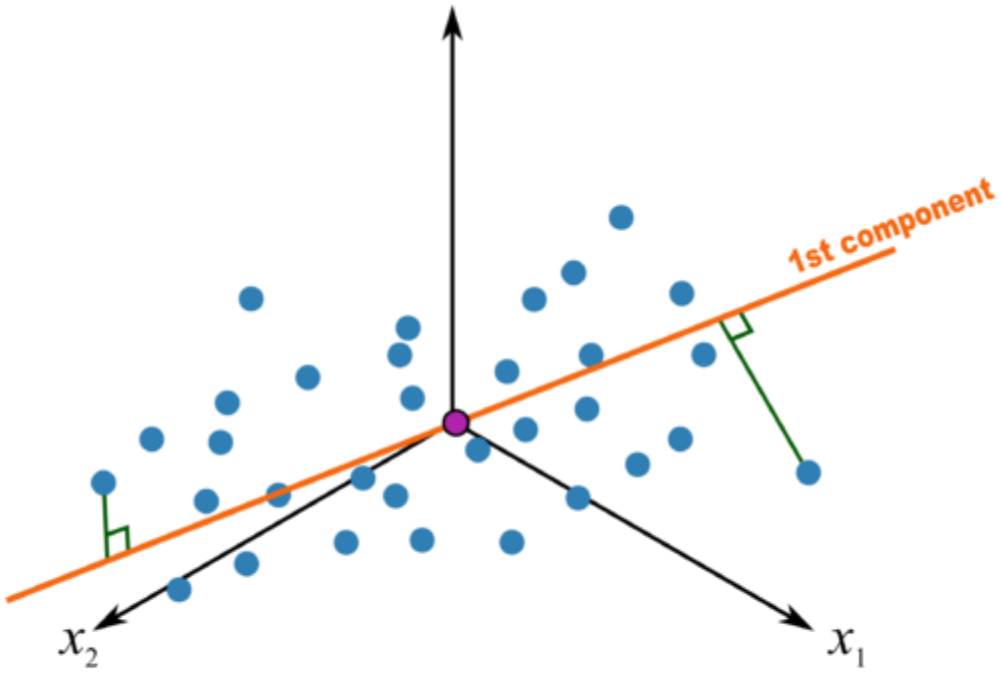
A powerful operation to reduce the number of variables in your data set, also known as “dimensionality reduction”. The number of variables is reduced in a way that as little information as possible is lost.

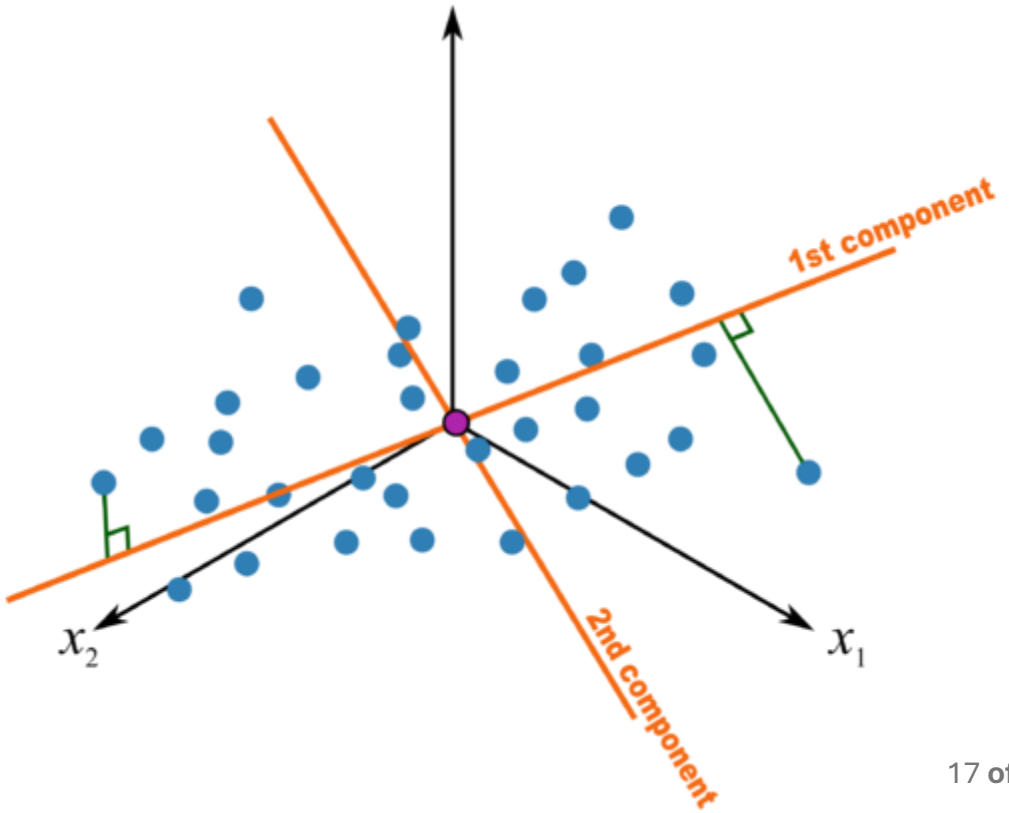




What is first principal component?

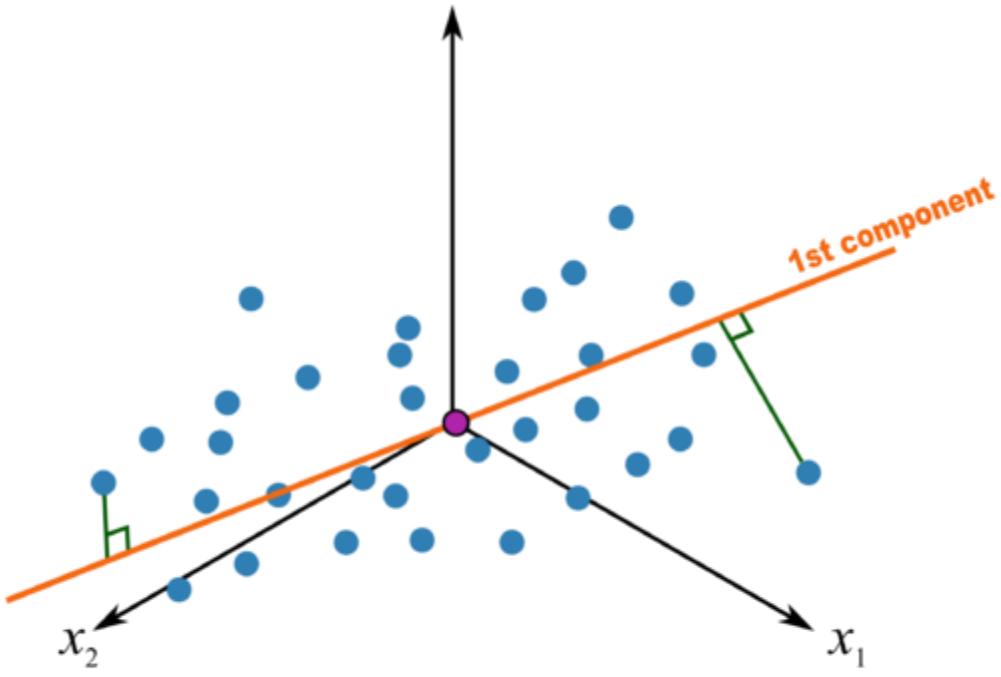
A vector/line pointing in the direction of greatest variability in the dataset. It is the direction where the data points are most spread out.

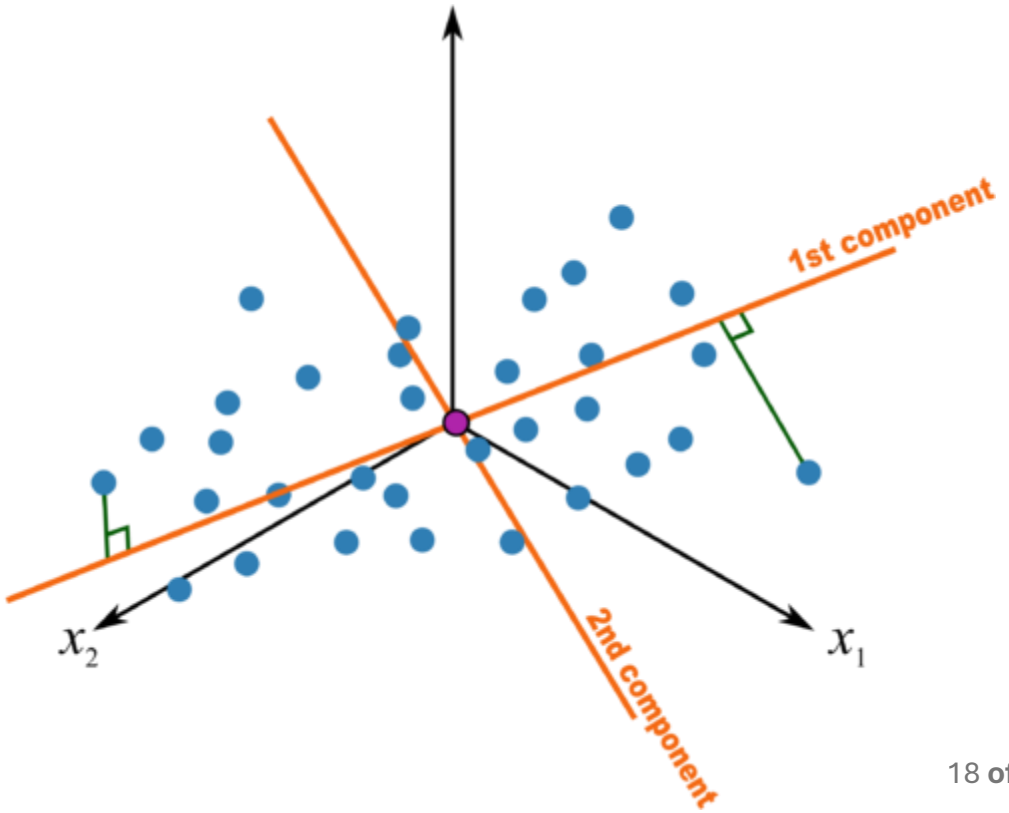




What is the second principal component?

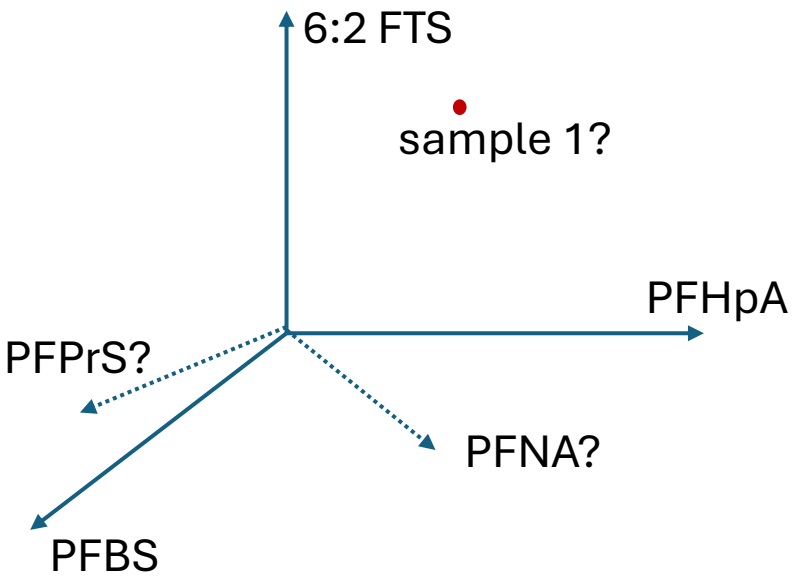
A vector/line pointing in the direction of second greatest variability in the dataset. It is normal to the first principal component.





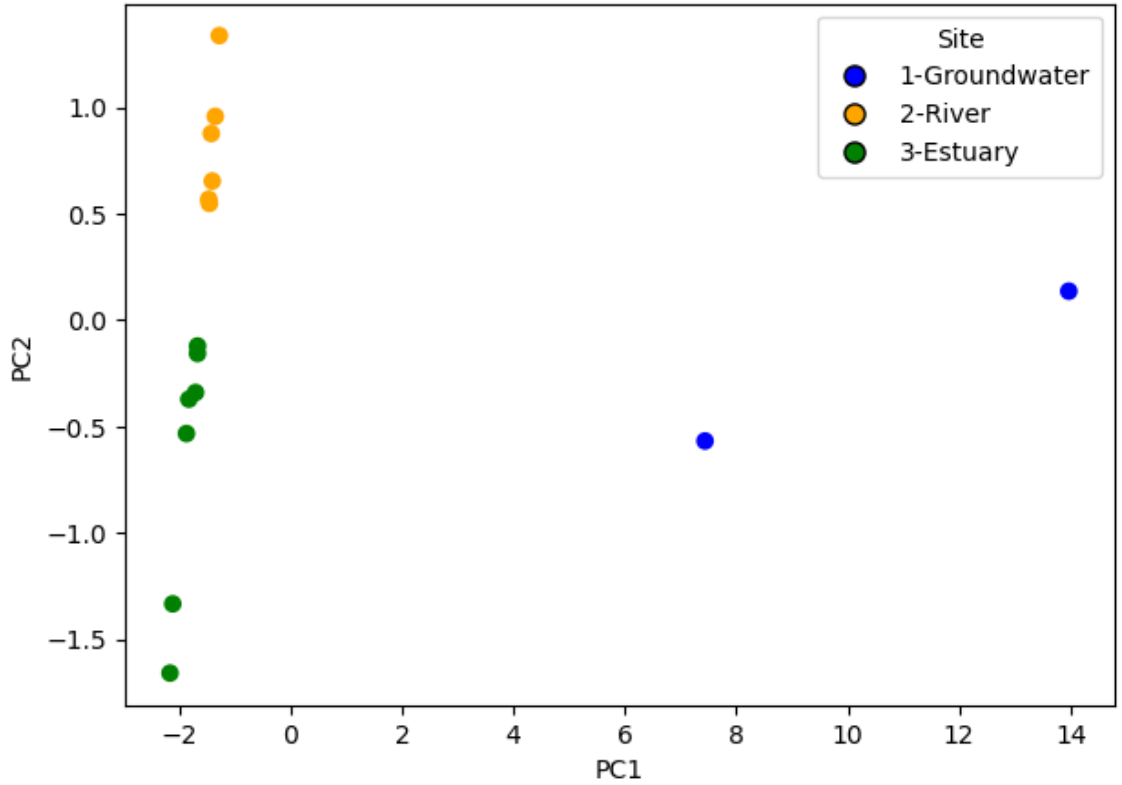
PCA for PFAS

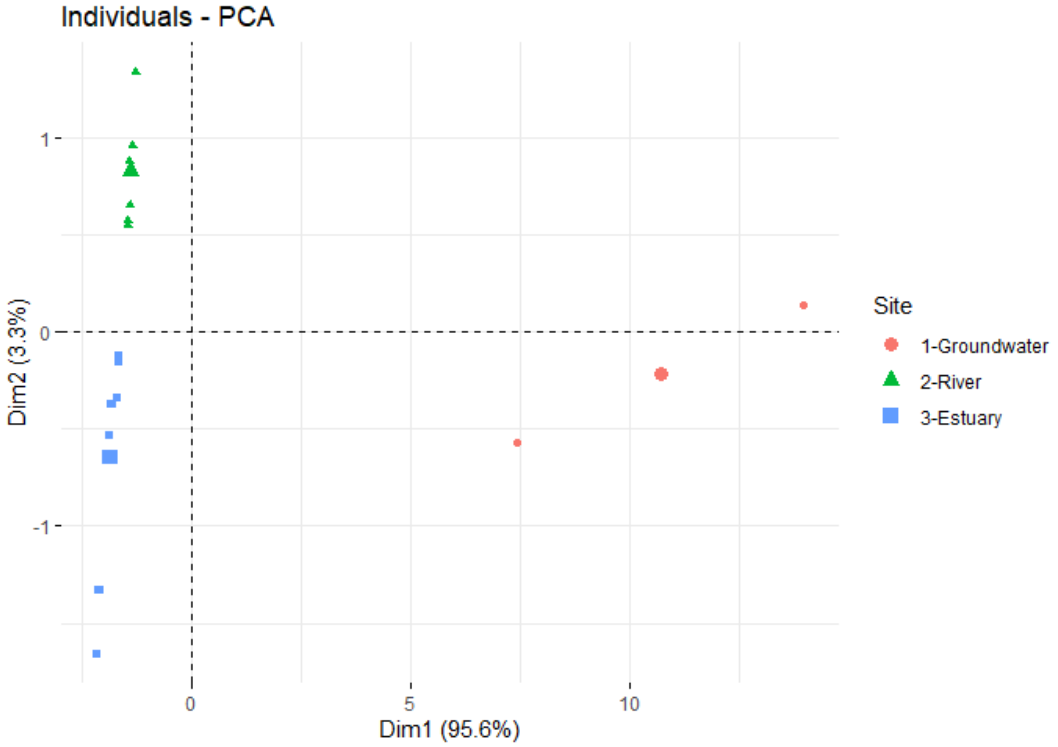
It is impossible to plot more than 3 dimensions, but we work with multi-dimensional data all the time.



$$\text{sample 1} = \begin{cases} 25.2\,\frac{ng}{mL}\ PFBS \\ 185.6\ \frac{ng}{mL}\,PFHpA \\ 99.5\ \frac{ng}{mL}\ 6{:}\,2\ FTS \\ 6.6\ \frac{ng}{mL}\,PFNA \\ 4.4\ \frac{ng}{mL}\,PFPrS \\ \ldots. \end{cases}$$

With PCA, you can plot PFAS results in 2D.





Plotting

```python
plt.figure()
plt.scatter(scores['PC1'], scores['PC2'], c=df['Site'].map(
    {'1-Groundwater': 'blue', '2-River': 'orange', '3-Estuary': 'green'}))
plt.xlabel('PC1')
plt.ylabel('PC2')

# Build legend handles from the color map
handles = [
    Line2D([0], [0], marker='o', color='w',
           label=label, markerfacecolor=col, markeredgecolor='k', markersize=8)
    for label, col in {'1-Groundwater': 'blue', '2-River': 'orange', '3-Estuary': 'green'}.items()
]
plt.legend(handles=handles, title='Site', loc='best', frameon=True)

plt.show()
```

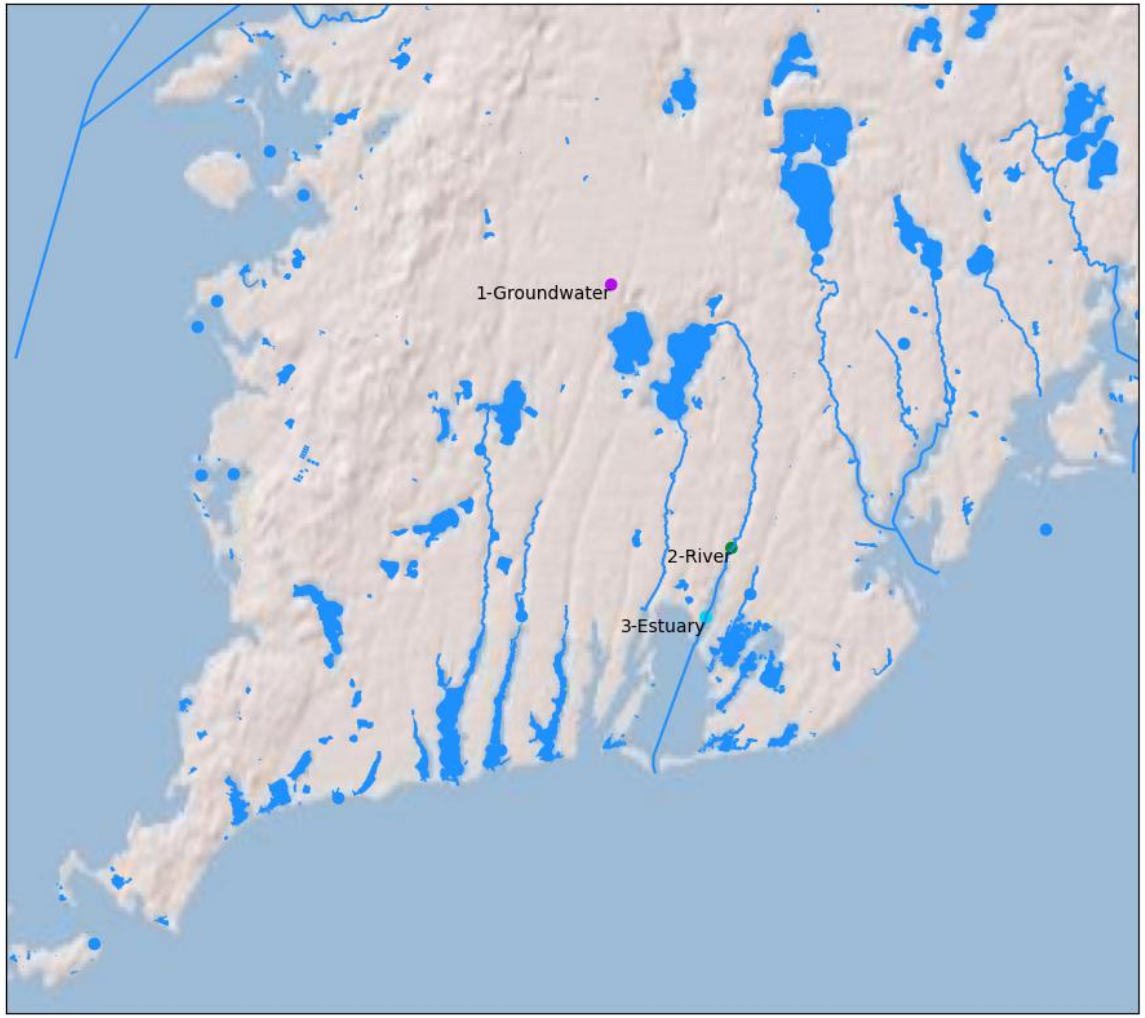
```r
# PCA
res.pca <- FactoMineR::PCA(
  df, quali.sup = which(names(df) == "Site"), graph = TRUE, ncp=2, scale=TRUE
  )
factoextra::fviz_pca_ind(res.pca, geom = 'point', habillage = 'Site')
```

Example data set

- Water samples from Matt Dunn (STEEP graduate) collected at various dates:
  - 2 groundwater samples
  - 6 river samples
  - 7 estuary samples
- Concentrations of 30 PFAS are reported for all samples

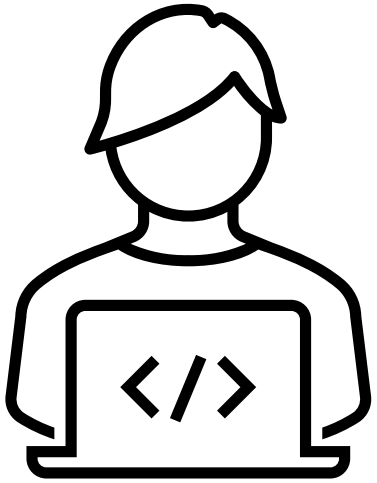
For PCA:

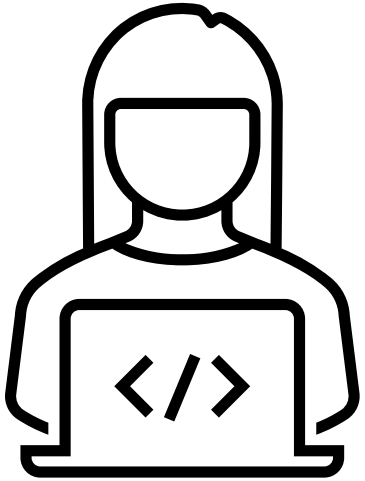
How are your samples described? What are your variables? How many dimensions do you expect?



Data pre-processing

- Load data
- Delete columns where less than 70% of data are available.
- Replace missing values or “<MDL”.
- In python:
  - Normalize data before PCA
- In R:
  - Make sure data type of all columns is numeric





Load and process data

```python
path = 'water_grab_samples_matt_dunn_cape_cod.csv'
df = pd.read_csv(path)
```

```r
path <- 'water_grab_samples_matt_dunn_cape_cod.csv'
df <- read.csv(path)
```

```python
# Count '<MDL' strings per column
mdl_count_per_col = df.eq('<MDL').sum()

# Remove columns with more than 30% '<MDL' values
threshold = 0.3 * len(df)
df = df.loc[:, mdl_count_per_col <= threshold]

# Replace remaining '<MDL' with 0
df.replace('<MDL', 0, inplace=True)

# drop 'Site' and Scale features
scaler = StandardScaler()
df_scaled = scaler.fit_transform(df.drop(columns=['Site']))
```

```r
# Count '<MDL' strings per column
mdl_count_per_col <- colSums(df == "<MDL")

# Remove columns with more than 30% '<MDL' values
threshold = 0.3 * nrow(df)
df = df[, mdl_count_per_col <= threshold]

# Replace remaining '<MDL' with 0
df[df == "<MDL"] <- 0
cols_to_convert <- c("PFPrS..ng.L.", "FOSA..ng.L.", "PFECHS..ng.L.")
df[cols_to_convert] <- lapply(df[cols_to_convert], as.numeric)
```

How does PCA work?

- Linear algebra!

- Easy to use packages in python:
  - import pandas as pd # to handle dataframes in python
  - import matplotlib.pyplot as plt # to create plots in python
  - from sklearn.decomposition import PCA # to perform PCA in python
  - from sklearn.preprocessing import StandardScaler

- Easy to use packages in R:
  - library(tidyverse)
  - library(FactoMineR)
  - library(factoextra)

Mini Exercise:

Install the packages/libraries in your python or R environment

How does PCA work?

Jupyter (Python)

```python
pca = PCA(n_components=2)
scores = pca.fit_transform(df_scaled)
scores = pd.DataFrame(scores, columns=['PC1', 'PC2'])
```

```python
explained = pca.explained_variance_ratio_
print('Explained variance ratio:', pca.explained_variance_ratio_)
```

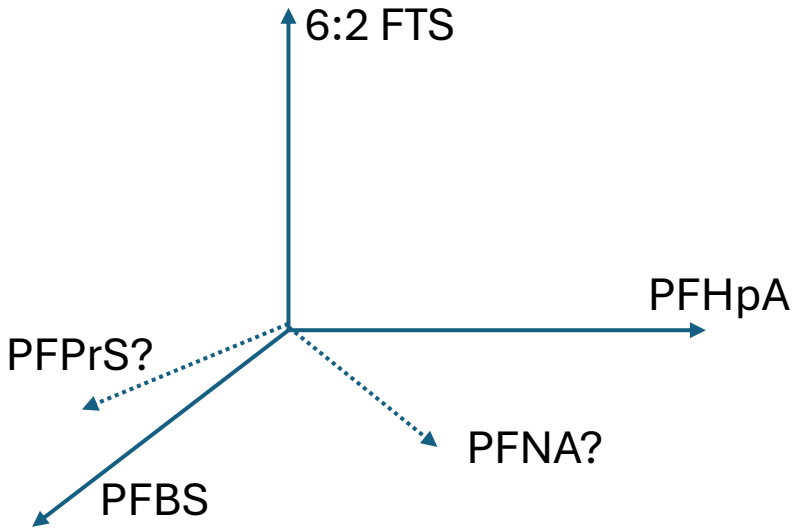
R

```r
# PCA
res.pca <- FactoMineR::PCA(
  df, quali.sup = which(names(df) == "Site"), graph = TRUE, ncp=2, scale=TRUE
  )
```

Principal Components (PC) for PFAS data

PC1: A vector/line pointing in the direction of greatest variability in the dataset.

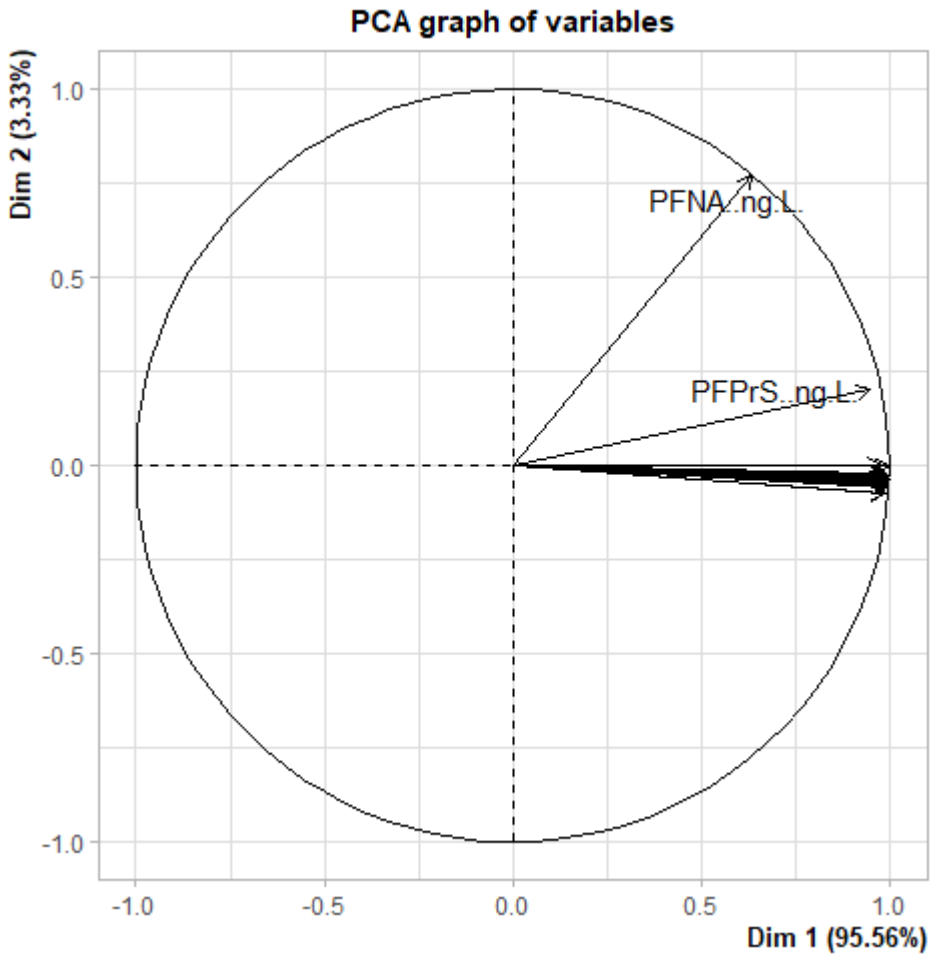
PC2: A vector/line pointing in the direction of second greatest variability in the dataset. It is normal to the first principal component.



$$\text{PC1} = \begin{cases} 0.23\ PFBS \\ 0.23\ PFHpA \\ 0.23\ 6{:}2\ FTS \\ 0.15\ PFNA \\ 0.22\ PFPrS \\ \ldots. \end{cases}, \text{PC2} = \begin{cases} 0.00\ PFBS \\ -0.05\ PFHpA \\ -0.05\ 6{:}2\ FTS \\ 0.94\ PFNA \\ 0.25\ PFPrS \\ \ldots. \end{cases},$$

How do you find out which PFAS have high influence on variability?

PFAS compounds in PCA coordinates (loadings)



$$\text{PC1} = \begin{cases} 0.23\ PFBS \\ 0.23\ PFHpA \\ 0.23\ 6{:}2\ FTS \\ 0.15\ PFNA \\ 0.22\ PFPrS \\ \ldots. \end{cases}, \text{PC2} = \begin{cases} 0.00\ PFBS \\ -0.05\ PFHpA \\ -0.05\ 6{:}2\ FTS \\ 0.94\ PFNA \\ 0.25\ PFPrS \\ \ldots. \end{cases}$$

$$\text{PFBS} = \begin{cases} 0.23 \\ 0 \end{cases}, \text{PFHpA} = \begin{cases} 0.23 \\ -0.05 \end{cases}, \text{6:2 FTS} = \begin{cases} 0.23 \\ -0.05 \end{cases},$$

$$\text{PFNA} = \begin{cases} 0.15 \\ 0.94 \end{cases}, \text{PFPrS} = \begin{cases} 0.22 \\ 0.25 \end{cases}$$

PC and Loadings

```python
# Loadings for first two PCs
loadings = pd.DataFrame(pca.components_, columns=df.drop(columns=['Site']).columns, index=['PC1', 'PC2'])
loadings = loadings.T  # Transpose for easier plotting
loadings.head(25)
```

```r
# Loadings
print(res.pca$var$coord)
```

Questions?